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[Chap 5] Data Visualization

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data visualization

- Extracting hidden meanings from data and using visual tools such as graphs Expressing

- The starting point of all analysis

- Well-made visualizations alone complete more than half of the data analysis
  - In the process of visualizing the meaning embedded in the data, the patterns and meanings of the data Becomes able to consider it sufficiently

- The 3 Basic Visualization Tools for Data Analysis
  - Histogram – Boxplot
  - Scatterplot

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[Case 1] Dasan Jeong Yak-yong

• In 1795, King Jeongjo inquired about the number of trees planted at Crown Prince Sado's tomb (Hyeonryungwon) and how many trees had been planted in surrounding villages – although carts filled with official documents (the Tree Planting Register),

no one was able to answer Jeongjo's questions.

King Jeongjo ordered Jeong Yak-yong to organize the number of trees.

• Jeong Yak-yong organized this into a table.

– Prepare eight sheets of paper for each of the eight districts – Create

a table on each sheet with year as the column heading and tree type as the row heading

• Based on this, a summary table was compiled by district (row) and year (column) – a total of

12,009,972 trees – it is known that

King Jeongjo greatly praised it and marveled at it for a long time.

• A case where exploitation was prevented by organizing the economic situation of each household into a table even when appointed as the Magistrate of Goksan.

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[Case 2] Nightingale's Rose Chart

- While operating a field hospital, witnessing situations where soldiers died from hospital infections rather than their injuries,

  Need to persuade home country for support to improve hospital hygiene conditions

- Developed a visualization tool to provide a clear overview of soldier deaths from hospital-acquired infections

  – Rose diagram

- From the chart, deaths due to hospital infections

  You can see at a glance that it is overwhelmingly more numerous

  than the injuries

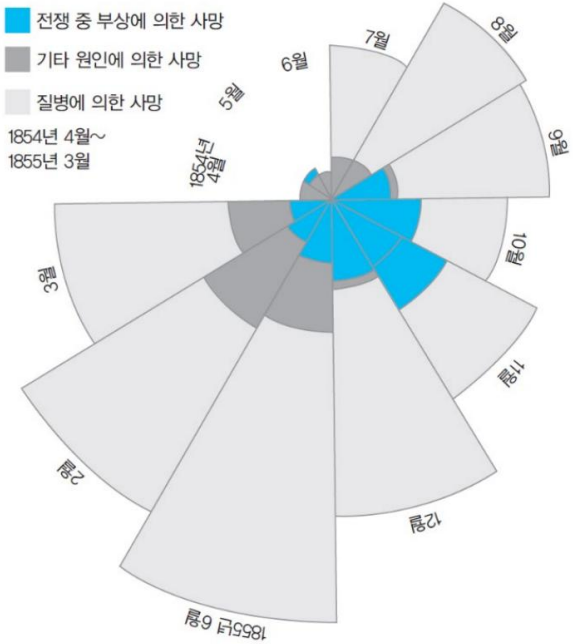
  themselves – through communication via visualization

  Saved the lives of many soldiers by securing support

  from the home country to improve the hospital environment.

- Considered a prime example of successful communication

  through visualization



[그림 5-2] 나이팅게일의 로즈 다이어그램 (자료 : https://www.uh.edu/engines/epi1712.htm)

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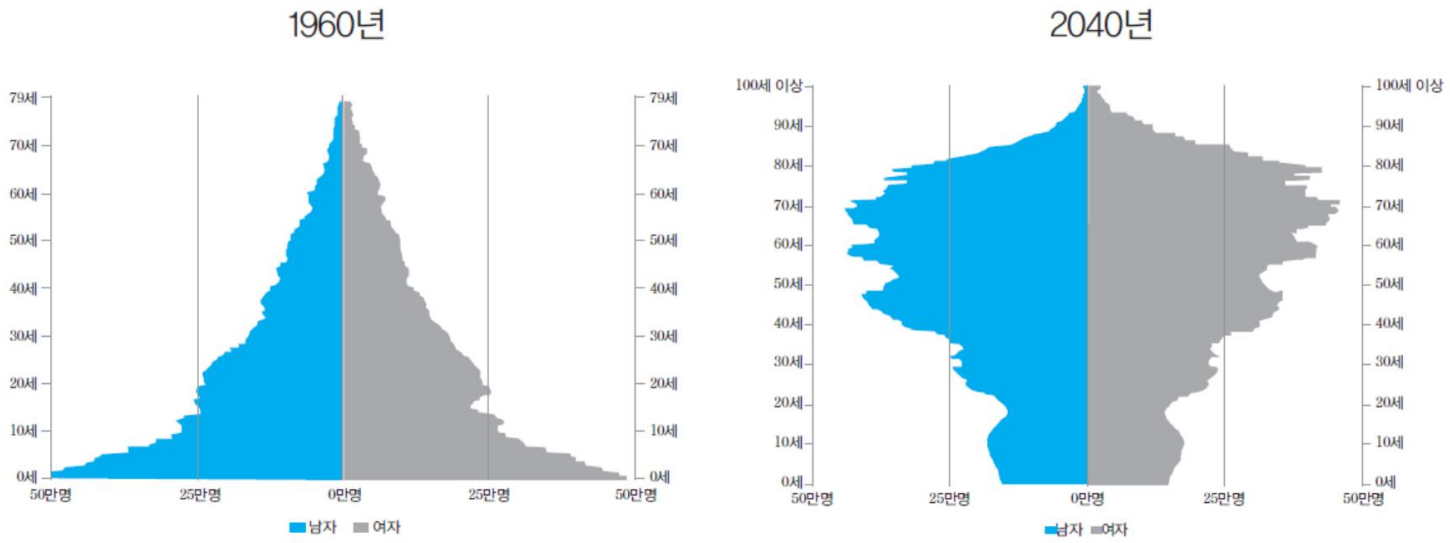
[Case 3] Population Pyramid

- Age is represented vertically, and the number of men and women in that age group is represented horizontally by the length of the horizontal line. histogram

- Confirmation of the formation of the baby boom generation due to high birth rates until the early 1970s

  – Caused social phenomena such as overcrowded classrooms in elementary, middle, and high schools, high barriers to university admission, and skyrocketing real estate prices

- Clearly visualizes the situation where low birth rates emerge as a national issue due to a sharp decline – Predicts the arrival of a super-aged society by 2040 if low birth rates persist – Foreshadows tremendous changes to society as a whole, as well as the economic and business environments, and provides important implications.



[그림 5–3] 인구 피라미드

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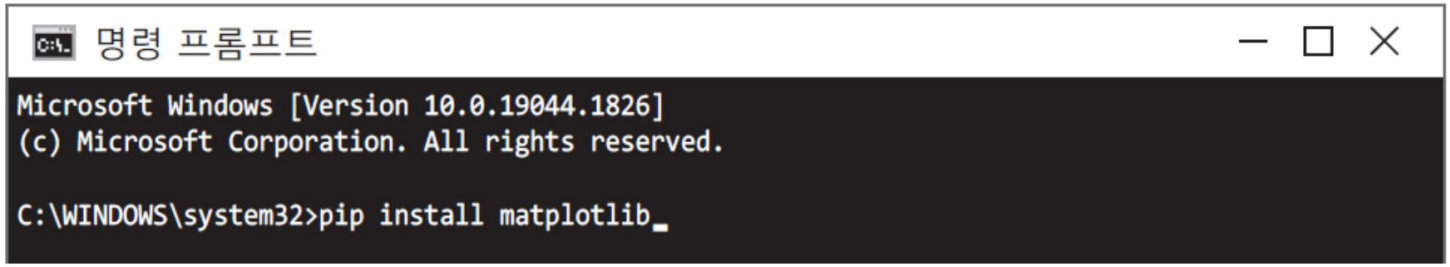
Basics of Data Visualization Programs

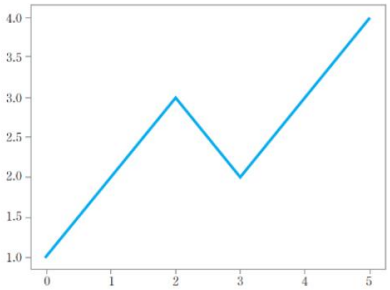
- Install matplotlib – Run

  cmd (Command Prompt) and run 'pip install matplotlib' (MacOS: pip3)



- Basic visualization examples

```
1  import matplotlib.pyplot as plt
2  x=[1,2,3,2,3,4]
3  plt.plot(x)
4  plt.show()
```



[그림 5-5] 기본적인 plot 함수의 출력

– (Line 1) Load matplitlib.pyplot with the short name plt – (Line 3) Prepare a graph of x values using the plot function – (Line 4)

Display the prepared graph on the screen using the show function

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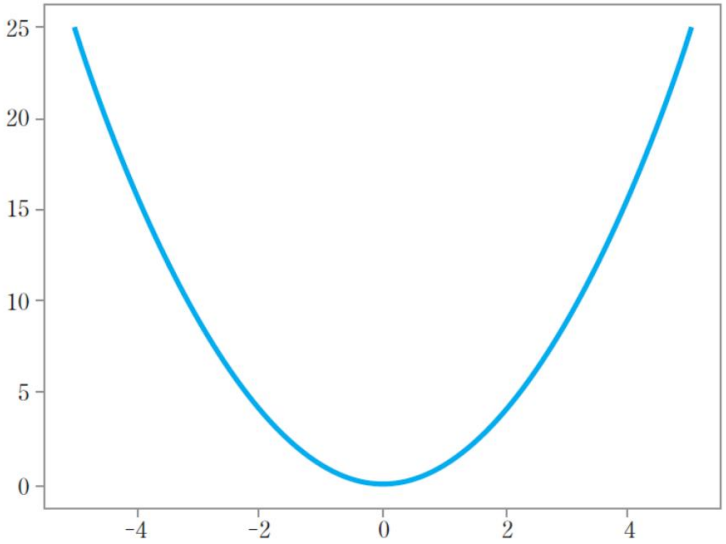
Basics of Programs for Data Visualization (Continued)

- Output graph of = 2

예제 ex5-1.py

```
1  import matplotlib.pyplot as plt
2  x=[-5, -4, -3, -2, -1, 0, 1, 2, 3 ,4, 5]
3  y=[]
4  for v in x:
5      y.append(v**2)
6  plt.plot(x,y)
7  plt.show()
```



[그림 5-6] plot을 이용한 이차함수 그래프

– (Line 4) Sequentially extract values from list x and assign them to v –

(Line 5) Sequentially append the squares of v to list y – (Line 6)

Prepare a graph connecting (x,y) in sequence using the plot function

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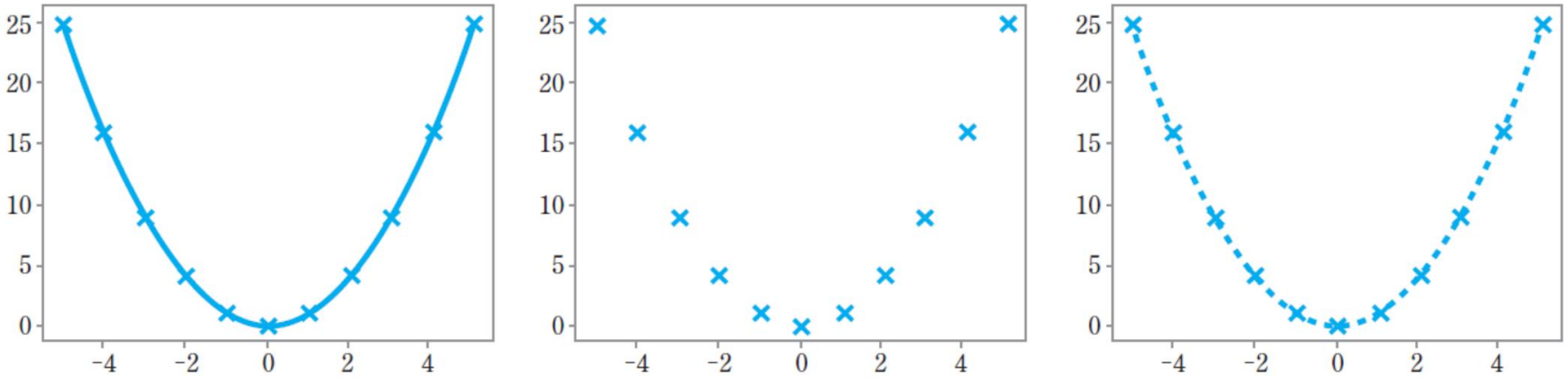
Basics of Programs for Data Visualization (Continued)

- plt.plot(x, y, marker='X') : Displays each point as an 'x'
- plt.plot(x, y, marker='X', linestyle='None') : Remove middle line
- plt.plot(x, y, marker='X', linestyle='--') : Change the line to a dotted line



[그림 5–7] plot에서 다양한 marker의 사용

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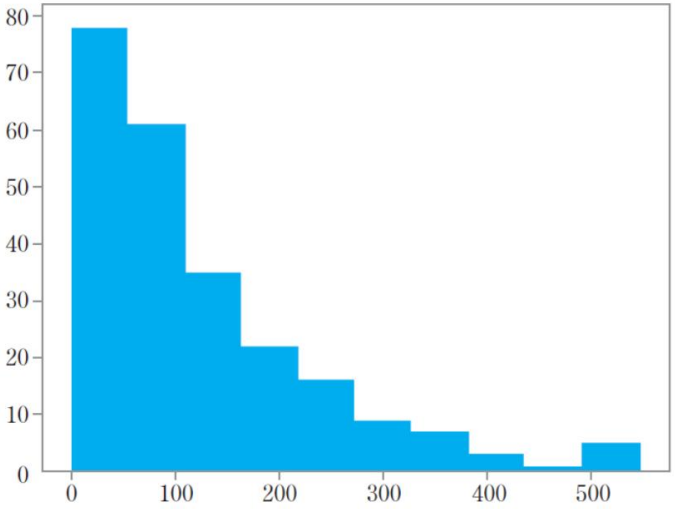
Data Visualization Practice: Histograms

- Histogram: Visualization of data distribution

- Output histogram of birthday gift prices from buylov.csv

예제 ex5-2.py

```
1 import pandas as pd
2 import matplotlib.pyplot as plt
3 b = pd.read_csv('buylov.csv')
4 bp = b['BGiftPrice']
5 plt.hist(bp)
6 plt.show()
```



[그림 5-8] 히스토그램 예시

– (Row 3) Read buylov.csv and save it to b – (Row 4) Extract only the 'BGiftPrice' column from b and save it to bp – (Row 5) Prepare a histogram for bp

- Futures prices are mostly distributed at lower price levels and show a long right tail – Corresponding to the fact that the average futures price was $124.6 and the median was $93.6 in Chapter 4.

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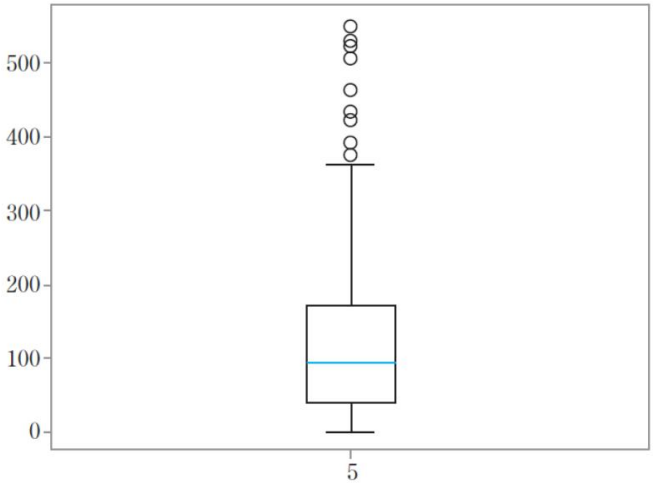
Data Visualization Practice: Box Plot

- Boxplot: Frequently used to compare the distributions of multiple values simultaneously
- Example of a boxplot on birthday gift prices

```
1  import pandas as pd
2  import matplotlib.pyplot as plt
3  b = pd.read_csv('buylov.csv')
4  bp = b['BGiftPrice']
5  plt.boxplot(bp)
6  plt.show()
```



[그림 5–9] 상자그림 예시

– (Line 3) Read buylov.csv and save it to b –

(Line 4) Extract only the 'BGiftPrice' column from b and save it

to bp – (Line 5) boxplot(bp): Prepare the values of bp as a box plot

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Data Visualization Practice: Box Plots (Continued)

- A box plot is a top-down view of a histogram – the high parts of the histogram are indicated by thick rectangles, and the low parts by lines.

- The thick line in the middle is the median

- The top and bottom boundaries of the box are the first and third quartiles of the data, respectively – The height range of the box contains the mainstream of the data (50% of the total data) – Box height: IQR (Inter-Quartile Range)

- The whiskers extend from the top and bottom edges of the box to 1.5 times the IQR – in the example, the bottom whisker appears short as it is truncated to the data minimum.

- Data outside the whiskers are suspicious outliers
  – Values outside the range of three times the IQR are strongly suspected outliers

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Data Visualization Practice: Box Plots (Continued)

- More useful when comparing multiple data

예제 ex5-3.py

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
b = pd.read_csv('buylov.csv')
bg = b[b['Role']=='Giver']
br = b[b['Role']=='Receiver']
print('<준  사람 선물가격>')
print('평균 :', np.mean(bg['BGiftPrice']))
print('중앙값:', np.median(bg['BGiftPrice']))
print('<받은 사람 선물가격>')
print('평균 :', np.mean(br['BGiftPrice']))
print('중앙값:', np.median(br['BGiftPrice']))
plt.boxplot([bg['BGiftPrice'],br['BGiftPrice']],labels=['Giver','Receiver'])
plt.show()
```

Separately insert data of the gift giver and receiver into bg and br.

Comparing the average and median of gift prices between givers and receivers—it is possible to interpret that recipients perceive the gift as having a higher value than its actual price.

Pass the two values bg['BGiftPrice'] and br['BGiftPrice'] to the boxplot to compare and output them as a box plot - box plot visualization shows the difference between the two values more clearly.

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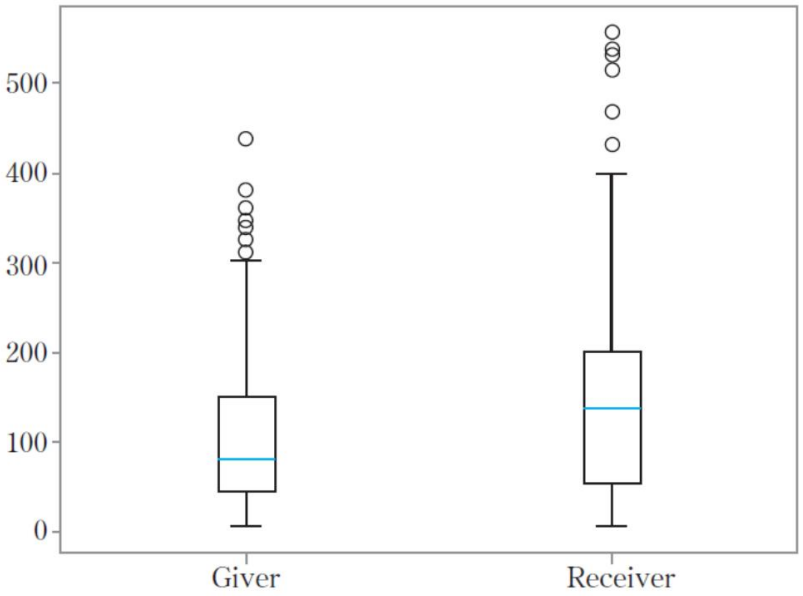
Data Visualization Practice: Box Plots (Continued)

- Output result of Example 5-3

```
<준  사람 선물가격>
평균 : 105.84328358208955
중앙값: 75.5
<받은 사람 선물가격>
평균 : 148.98058252427185
중앙값: 133.0
```



[그림 5-10] 상자그림을 이용한 데이터 비교

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Data Visualization Practice: Scatter Plot

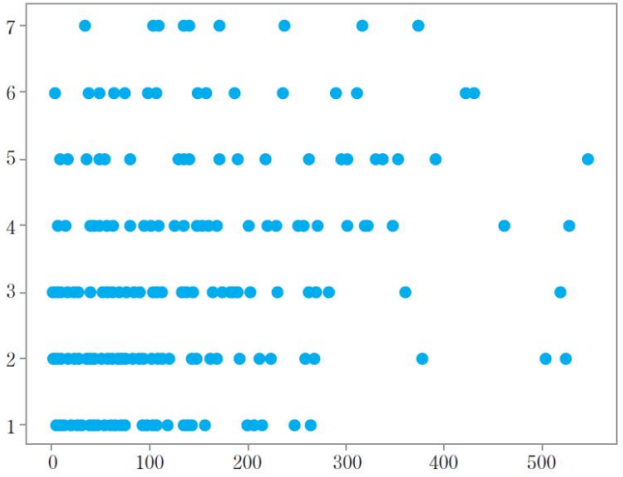
- Scatterplot: Visualizes the relationship between two data series

- Analyze the relationship between gift price and the degree of gratitude

```
1  import pandas as pd
2  import matplotlib.pyplot as plt
3  b = pd.read_csv('buylov.csv')
4  plt.scatter(b['BGiftPrice'], b['Apprec-7'])
5  plt.show()
```

  – (Row 4) Prepare a scatter plot with futures prices on the x-axis and gratitude levels on the y-axis.

- No clear pattern is visible between futures prices and appreciation in the output scatter plot



[그림 5-11] 산점도 예시

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Data Visualization Practice: Scatter Plot (Continued)

• Diagram the data of the gift giver and receiver by separating them into numbers 1 and 2.

예제 ex5-4.py

```
1  import pandas as pd
2  import matplotlib.pyplot as plt
3  b = pd.read_csv('buylov.csv')
4  bg = b[b['Role']=='Giver']
5  br = b[b['Role']=='Receiver']
6  plt.figure(1)
7  plt.scatter(bg['BGiftPrice'], bg['Apprec-7'])
8  plt.figure(2)
9  plt.scatter(br['BGiftPrice'], br['Apprec-7'])
10 plt.show()
```

– (Lines 4–5) Separate data for gift givers and receivers into bg and br

– (Lines 6–7) Set figure number to 1 and in bg

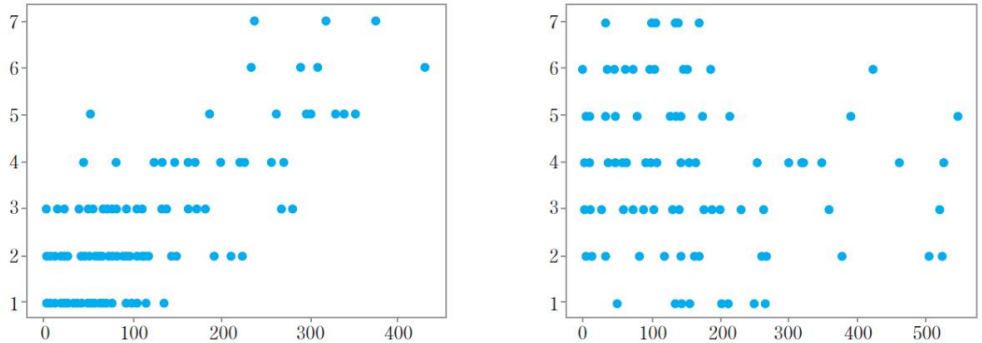
Prepare a scatter plot of gift prices and gratitude levels.

– (Lines 8–9) Set figure number to 2 and in br

Prepare scatter plots of futures prices and gratitude levels – (Row 10) Display figures 1 and 2 on the screen simultaneously

• High correlation observed for the gift giver, no correlation seen for the receiver –

While gift givers likely expected the recipient to appreciate the expensive gift, it was observed that for the receivers, there was no correlation between price and the degree of gratitude.



(a) 선물을 준 사람의 데이터 (b) 선물을 받은 사람의 데이터

[그림 5-12] 역할에 따라 구분된 산점도